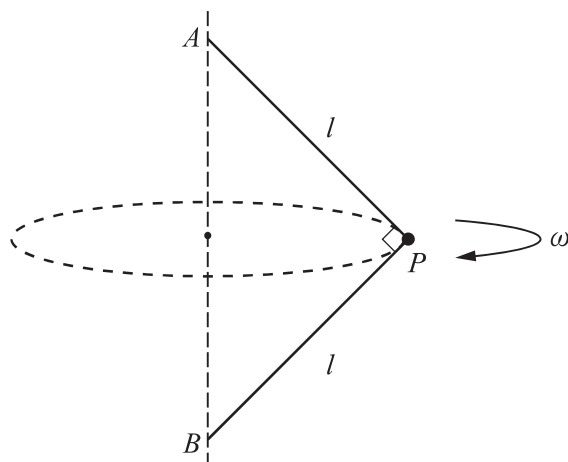




- 1** A particle P of mass m is attached to two light inextensible strings each of length l . The end of one string is attached to a fixed point A and the end of the other string is attached to a fixed point B , with A vertically above B . Angle APB is a right angle. The particle P rotates in a horizontal circle at a constant angular speed ω with both strings taut (see diagram).



Find the tension in string AP in terms of m , g , l and ω .

[4]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the page.